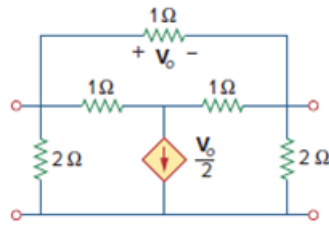


Problem

Using *PSpice* or *MultiSim*, find the transmission parameters for the network in Fig. 19.126.

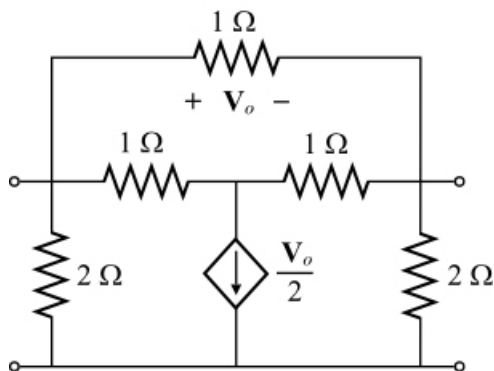
Figure 19.126



Step-by-step solution

Step 1 of 16

Consider the following circuit diagram:



Step 2 of 16

Figure 1

Step 3 of 16

Write the general transmission parameters.

$$\mathbf{A} = \left. \frac{\mathbf{V}_1}{\mathbf{V}_2} \right|_{\mathbf{I}_2=0} \quad \dots (1)$$

$$\mathbf{B} = - \left. \frac{\mathbf{V}_1}{\mathbf{I}_2} \right|_{\mathbf{V}_2=0} \quad \dots (2)$$

$$\mathbf{C} = \left. \frac{\mathbf{I}_1}{\mathbf{V}_2} \right|_{\mathbf{I}_2=0} \quad \dots (3)$$

$$\mathbf{D} = - \left. \frac{\mathbf{I}_1}{\mathbf{I}_2} \right|_{\mathbf{V}_2=0} \quad \dots (4)$$

Step 4 of 16

To calculate the values of **A** and **C** parameters, connect a test voltage source to input port and open circuit the output port. Draw the modified circuit diagram.

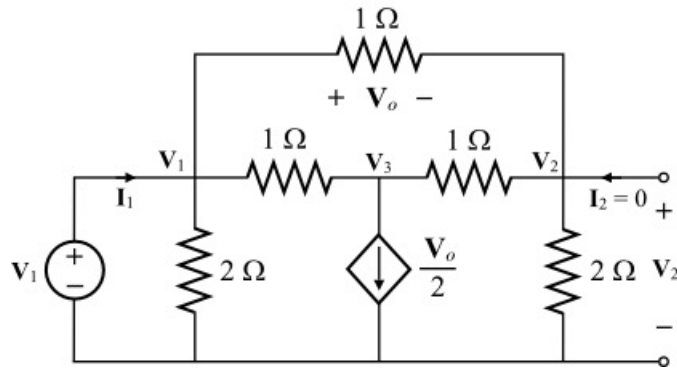


Figure 2

Step 5 of 16

From the Figure 2,

$$V_o = V_1 - V_2$$

Apply Kirchhoff's current law at node V_2 .

$$\frac{V_2 - V_1}{1} + \frac{V_2}{2} + \frac{V_2 - V_3}{1} = 0$$

$$-V_1 + (1 + 0.5 + 1)V_2 - V_3 = 0$$

$$-V_1 + 2.5V_2 - V_3 = 0$$

$$V_3 = 2.5V_2 - V_1 \dots\dots (5)$$

Step 6 of 16

Apply Kirchhoff's current law at node V_3 .

$$\begin{aligned}\frac{V_3 - V_1}{1} + \frac{V_3 - V_2}{1} + \frac{V_o}{2} &= 0 \\ \frac{V_3 - V_1}{1} + \frac{V_3 - V_2}{1} + \frac{V_1 - V_2}{2} &= 0 \\ (-1 + 0.5)V_1 - (1 + 0.5)V_2 + (1 + 1)V_3 &= 0 \\ -0.5V_1 - 1.5V_2 + 2V_3 &= 0\end{aligned}$$

Substitute $2.5V_2 - V_1$ for V_3 in the above equation.

$$\begin{aligned}-0.5V_1 - 1.5V_2 + 2(2.5V_2 - V_1) &= 0 \\ -0.5V_1 - 1.5V_2 + 5V_2 - 2V_1 &= 0 \\ 2.5V_1 &= 3.5V_2 \\ V_1 &= \frac{3.5V_2}{2.5}\end{aligned}$$

Apply Kirchhoff's current law at node V_1 .

$$\begin{aligned}I_1 &= \frac{V_1}{2} + \frac{V_1 - V_2}{1} + \frac{V_1 - V_3}{1} \\ &= (0.5 + 1 + 1)V_1 - V_2 - V_3 \\ &= 2.5V_1 - V_2 - V_3\end{aligned}$$

Substitute $2.5V_2 - V_1$ for V_3 in the above equation.

$$\begin{aligned}I_1 &= 2.5V_1 - V_2 - 2.5V_2 + V_1 \\ &= 3.5V_1 - 3.5V_2 \\ &= 3.5\left(\frac{3.5V_2}{2.5}\right) - 3.5V_2 \\ &= \frac{12.25V_2 - 8.75V_2}{2.5} \\ &= \frac{3.5V_2}{2.5}\end{aligned}$$

Step 7 of 16

Substitute $\frac{3.5V_2}{2.5}$ for V_1 in the equation (1).

$$\begin{aligned}A &= \frac{V_1}{V_2} \\ &= \frac{\frac{3.5V_2}{2.5}}{V_2} \\ &= \frac{3.5}{2.5} \\ &= 1.4\end{aligned}$$

Substitute $\frac{3.5V_2}{2.5}$ for V_1 in the equation (3).

$$\begin{aligned}C &= \frac{I_1}{V_2} \\ &= \frac{\frac{3.5V_2}{2.5}}{V_2} \\ &= \frac{3.5}{2.5} \\ &= 1.4\text{ S}\end{aligned}$$

Step 8 of 16

To calculate the values of **B** and **D** parameters, connect a test current source to input port and short circuit the output port. Draw the modified circuit diagram.

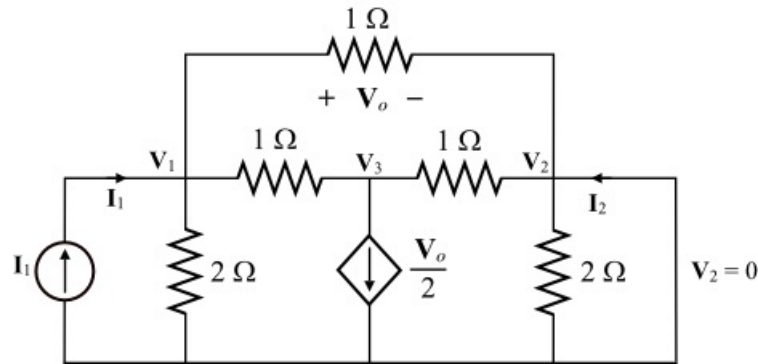


Figure 3

Step 9 of 16

From the Figure 3,

$$V_o = V_1$$

Apply Kirchhoff's current law at node V_2 .

$$I_2 = \frac{-V_1}{1} + \frac{-V_3}{1}$$

$$I_2 = -V_1 - V_3 \dots\dots (6)$$

Apply Kirchhoff's current law at node V_3 .

$$\frac{V_3 - V_1}{1} + \frac{V_3}{1} + \frac{V_o}{2} = 0$$

$$\frac{V_3 - V_1}{1} + \frac{V_3}{1} + \frac{V_1}{2} = 0$$

$$(-1 + 0.5)V_1 + (1 + 1)V_3 = 0$$

$$2V_3 = 0.5V_1$$

$$V_3 = \frac{0.5V_1}{2} \dots\dots (7)$$

Step 10 of 16

Substitute $\frac{0.5V_1}{2}$ for V_3 in the equation (6).

$$\begin{aligned} I_2 &= -V_1 - \frac{0.5V_1}{2} \\ &= \frac{-2V_1 - 0.5V_1}{2} \\ &= \frac{-2.5V_1}{2} \end{aligned}$$

Apply Kirchhoff's current law at node V_1 .

$$\begin{aligned} I_1 &= \frac{V_1}{2} + \frac{V_1}{1} + \frac{V_1 - V_3}{1} \\ &= (0.5 + 1 + 1)V_1 - V_3 \\ &= 2.5V_1 - V_3 \end{aligned}$$

Substitute $\frac{0.5V_1}{2}$ for V_3 in the above equation.

$$\begin{aligned} I_1 &= 2.5V_1 - \frac{0.5V_1}{2} \\ &= \frac{5V_1 - 0.5V_1}{2} \\ &= \frac{4.5V_1}{2} \end{aligned}$$

Step 11 of 16

Substitute $\frac{-2.5V_1}{2}$ for I_2 in the equation (2).

$$\begin{aligned} B &= -\frac{V_1}{I_2} \\ &= -\frac{V_1}{\frac{-2.5V_1}{2}} \\ &= \frac{2}{2.5} \\ &= 0.8 \Omega \end{aligned}$$

Substitute $\frac{3.5V_2}{2.5}$ for I_1 in the equation (4).

$$\begin{aligned} D &= -\frac{I_1}{I_2} \\ &= -\frac{\frac{4.5V_1}{2}}{\frac{-2.5V_1}{2}} \\ &= \frac{4.5}{2.5} \\ &= 1.8 \end{aligned}$$

Therefore, the transmission parameters, $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$ are $\begin{bmatrix} 1.4 & 0.8 \Omega \\ 1.4 \text{ S} & 1.8 \end{bmatrix}$.

Step 12 of 16

PSpice analysis:

- Open the CIS Demo software.
- Once you are in the capture session frame, click on the menu item File, select new, and then click on Project.
- In the New Project box, type 'selected file name' in the Name text box. Ensure that the Analog or Mixed-signal circuit Wizard is activated.
- Connected the circuit diagram as per the Circuit.
- Double click on each component and enter the given values.
- Click the New Simulation Profile icon and enter a name in the Name text box. You will need to enter the appropriate settings for this project in the Simulation
- Setting box. Click the Analysis tab, and select Bias Point from the Analysis type list. Click OK and save the document.
- Click on the run icon. You will observe the A/D Demo screen. When you close this screen, you will observe the bias voltages and currents.

To calculate the values of **A** and **C** parameters, connect a 1 V test voltage source to input port and open circuit the output port.

The following is the PSpice schematic:

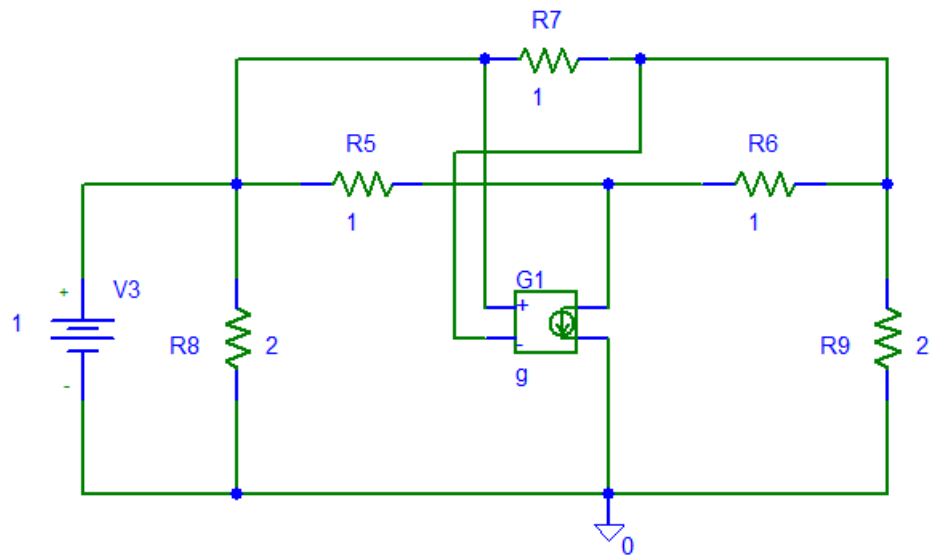


Figure 4

Run the Schematic, the following is the schematic circuit with voltage and current values.

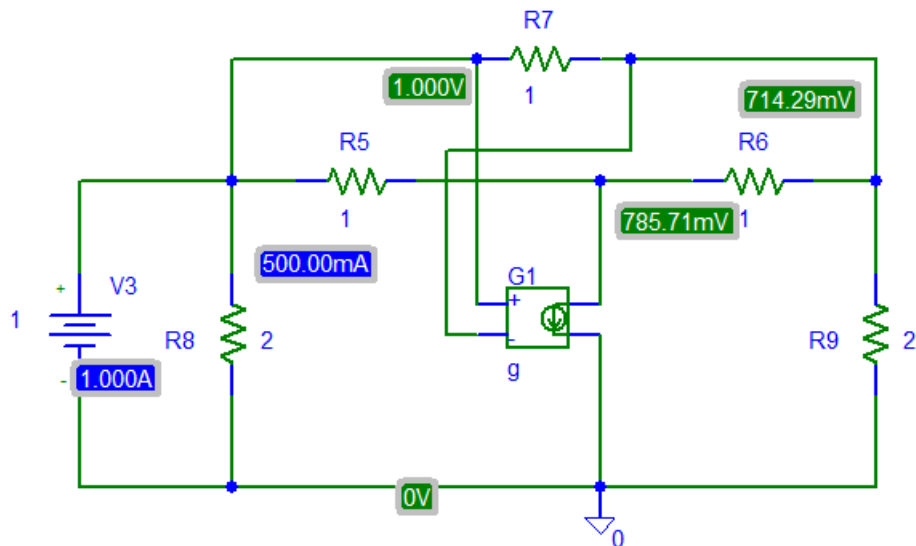


Figure 5

From the Figure 5,

$$I_1 = 1 \text{ A}$$

$$I_2 = 0 \text{ A}$$

$$V_1 = 1 \text{ V}$$

$$V_2 = 714.29 \text{ mV}$$

Substitute 1 V for V_1 and 714.29 mV for V_2 in the equation (1).

$$\begin{aligned} A &= \frac{V_1}{V_2} \\ &= \frac{1}{714.29 \times 10^{-3}} \\ &= 1.4 \end{aligned}$$

Substitute 1 A for I_1 and 714.29 mV for V_2 in the equation (3).

$$\begin{aligned} C &= \frac{I_1}{V_2} \\ &= \frac{1}{714.29 \times 10^{-3}} \\ &= 1.4 \text{ S} \end{aligned}$$

Step 15 of 16

To calculate the values of **B** and **D** parameters, connect a 1 A test current source to input port and short circuit the output port.

The following is the PSpice schematic:

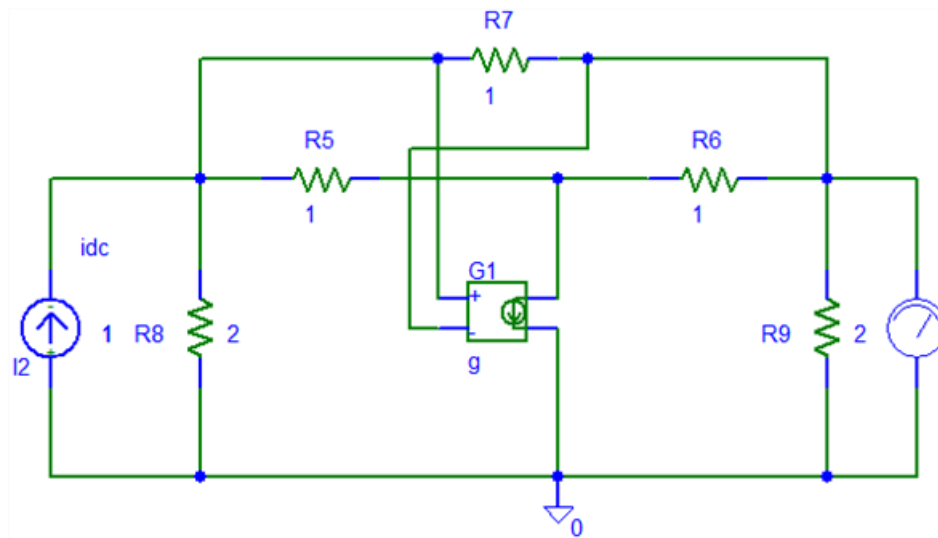


Figure 6

Step 16 of 16

Run the Schematic, the following is the schematic circuit with voltage and current values.

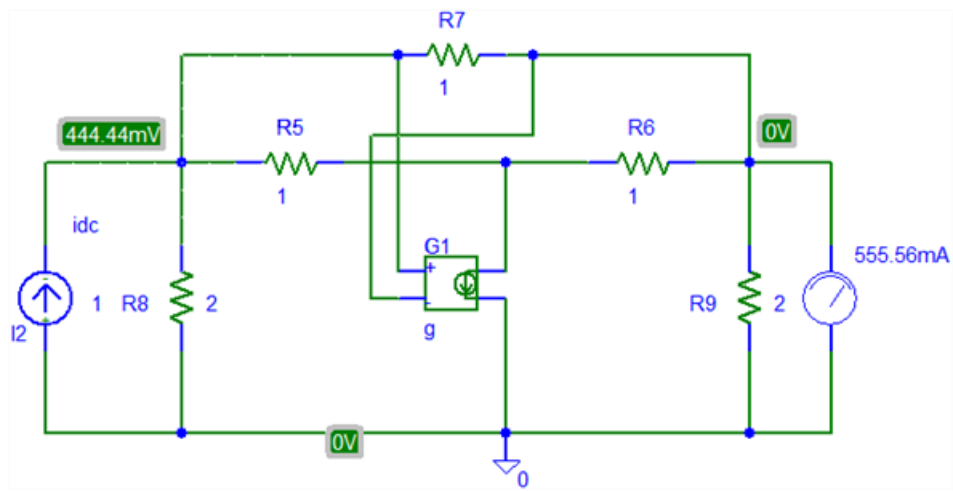


Figure 7

From the Figure 7,

$$I_1 = 1 \text{ A}$$

$$I_2 = -555.56 \text{ mA}$$

$$V_1 = 444.44 \text{ mV}$$

$$V_2 = 0 \text{ V}$$

Substitute 444.44 mV for V_1 and -555.56 mA for I_2 in the equation (2).

$$\begin{aligned} B &= -\frac{V_1}{I_2} \\ &= -\frac{444.44 \times 10^{-3}}{-555.56 \times 10^{-3}} \\ &= 0.8 \Omega \end{aligned}$$

Substitute 1 A for I_1 and -555.56 mA for I_2 in the equation (4).

$$\begin{aligned} D &= -\frac{I_1}{I_2} \\ &= -\frac{1}{-555.56 \text{ mA}} \\ &= 1.8 \end{aligned}$$

Therefore, the transmission parameters, $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$ are $\begin{bmatrix} 1.4 & 0.8 \Omega \\ 1.4 \text{ S} & 1.8 \end{bmatrix}$.